Ethan Edwards

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EDUCATION

University of Virginia May 2028
B.A. in Computer Science GPA: 3.79 / 4.00 | Major GPA: 3.95 / 4.00

EXPERIENCE

Software Executive – Robotics & Autonomy January 2026 – Present
Mechatronics and Robotics Society (MARS), UVA marsatuva.org

- Lead **10+ software developers** within a **140+ member** robotics organization, directing development and integrating contributions across autonomy, perception, navigation, and control.
- Served as the **primary autonomy-stack point of contact** before becoming Software Executive, contributing autonomous systems and on-site mission control to first-place NASA Lunabotics and third-place international Lunabotics finishes.
- Architected and implemented the ROS 2 state machine and autonomous excavation/dumping systems, coordinating asynchronous ROS 2 Actions for robotic operations.
- Integrated RTAB-Map visual SLAM, Nav2 navigation with A* and Theta* planning, and AprilTag localization; coordinated LiDAR/depth-camera point clouds and PyTorch U-Net segmentation for perception.
- Developed C++ socket-based telemetry connecting robot state and sensor feedback to a real-time React operator interface.

IT Developer June 2025 – August 2026
Trusted Care Foundation Fairfax, VA

- Built and deployed a production website serving **100+ monthly users** for medical-training program registration; refined the site through stakeholder feedback.
- Integrated registration and CRM data workflows, saving upper management **3+ hours every two weeks** and improving access to organizational information.

Machine Learning Intern July 2026 – August 2026
LGI Tech Leesburg, VA

- Developed an LLM-assisted transcription refinement pipeline using Python and prompt engineering to parse structured JSON output and apply corrections.
- Measured **false-positive and true-positive rates** to evaluate transcription corrections and assess the pipeline's accuracy and reliability.

PROJECTS

Hoos Taking Notes Fall 2026
Full-Stack Software Engineering | *Scrum Lead*

- Developing a collaborative note-sharing platform deployed on Heroku using Django, PostgreSQL, and Amazon S3 for persistent storage and uploaded course materials.
- Implementing role-based access control and Google authentication to support secure note sharing.
- Serving as Scrum Lead for a **five-person team**, coordinating weekly reviews, development tasks, and deadlines.

TOP Compiler Extensions Fall 2026
Compiler Design | *Academic Project*

- Extended TOP with for-loops, lambda expressions, and sequences across the compiler pipeline, from parsing and frontend processing through code generation.
- Designed and documented extension architecture; updated the type checker and ownership/memory-safety analysis to support new constructs while preserving type and memory safety.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, C#, JavaScript/TypeScript, SQL, HTML/CSS, x86 Assembly
Frameworks & Tools: ROS 2, Nav2, SLAM, PyTorch, Django, React, Linux, Git, Docker
Technical Areas: SLAM, Computer Vision, Path Planning, Compiler Design, Software Engineering
Relevant Coursework: Data Structures and Algorithms, Discrete Math, Computer Systems and Organization